Multimodal Guided Image Editing for E-Commerce Catalog Enrichment

Rohit Doriya
rohitdoriya@iisc.ac.in

Abstract

The e-commerce industry faces a critical challenge: generating diverse, high-quality product imagery at scale while maintaining strict fidelity to product identity. Current generative AI tools lack precise multimodal control and often corrupt fine-grained details such as logos, text, and product geometry. We propose a three-stage framework for multimodal guided image editing that directly addresses e-commerce requirements. **Stage 1** generates pseudo-paired training data by inverting established transformations on existing product assets, solving the industry-wide scarcity of paired datasets. **Stage 2** introduces disentangled multimodal condition blending via lightweight adapters that inject text, reference images, and spatial cues into the diffusion model's modulation space. **Stage 3** employs a council-of-experts approach with VLM-based reward signals to enforce product fidelity, realism, and instruction adherence. This three-stage pipeline is designed to enable high-fidelity, realistic edits that preserve brand identity while reducing manual post-production overhead.

Figure 1: Representative scene-enriched image demonstrating the conversion of a studio product image into a realistic lifestyle and interaction scene.

1 Introduction

E-commerce teams start from studio shots: clean product images that preserve brand and product details. Those images then need to become lifestyle images, advertising creatives, and short video frames. Manual post-production is slow, while prompt-only generation can change the product itself.

Scene enrichment is therefore a controlled editing problem. The product image is not only inspiration; it is a strict constraint. A useful system must preserve logos, text, shape, color, texture, and scale while changing context, lighting, composition, and human interaction.

1.1 Problem Statement

The core challenge is controllable editing under strict fidelity constraints. Existing image editors can make visually attractive scenes, but catalog use fails if the edit changes brand text, product geometry, material texture, or object affordance. We identified several recurring, non-exclusive failure modes, which are summarized in Table 1.

Table 1: Non-exclusive failure labels observed

Failure Label	Observed Rate (%)
Unnatural hand-object interaction	44%
Product identity drift	36%
Text/logo corruption	31%
Hallucinated accessories/parts	27%
Scale or perspective mismatch	22%

2 Proposed Task

The proposed task is **product-preserving scene enrichment**. Given a product image, a natural-language instruction, and optional multimodal conditions such as a reference image or sparse spatial cue, the model generates a context-rich image.

- **Input:** Studio/product image plus text instruction.
- **Optional controls:** Reference context image, mask, crop, outpainting region, or sparse spatial hint.
- **Output:** Lifestyle, campaign, or interaction image.
- **Acceptance criteria:** Identity retained, prompt satisfied, realistic scene, valid affordance.

3 Methodology

The method follows the proposal's three stage design. Stage 1 formulates data creation as an inverse problem. Stage 2 blends heterogeneous conditions such as text, reference imagery, and spatial hints. Stage 3 evaluates outputs with expert style scores for fidelity, instruction adherence, realism, and affordance.

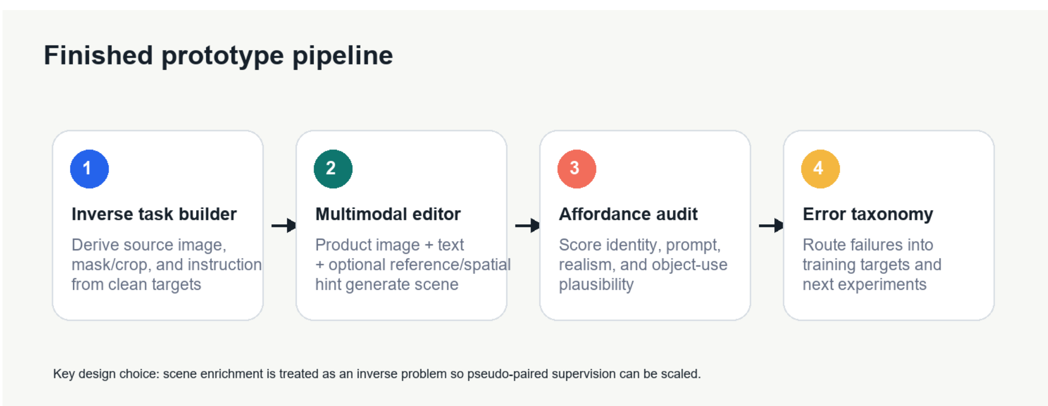

Figure 2: Finished prototype pipeline.

Evaluation Rubric To accurately judge the outputs, we established a four-dimensional evaluation rubric:

- **Product identity:** Text/logo retention, color, shape, texture, and geometry.

- **Prompt following:** Scene, activity, object placement, and requested attributes.
- **Perceptual realism:** Lighting, artifacts, perspective, and composition.
- **Affordance plausibility:** Contact, grip, scale, hand pose, and interaction target.

4 Experiments and Implementation Details

The source work explored an MLE product image collection. After duplicate product IDs were dropped, the working notes recorded 110,823 unique images across 307 item groups, averaging roughly 360 images per item. For the final prototype, the evaluation was scoped to 15 products and 30 well-posed enrichment prompts.

- **Data curation:** Duplicate FSN removal, category filtering, pilot split design.
- **Categories tested:** Mobile/tablet, appliance/UPS, laptop accessory, beauty tool, tripod/selfie stick, and small electronics.
- **Baselines:** FireRed-Image-Edit, LongCat-Image-Edit, and Nano Banana Pro. (Note: Flux and Qwen-Image-Edit were excluded from the final comparison after inference failures under available compute constraints).
- **Scored outputs:** 90 generated outputs across 30 prompts and 3 model families.

4.1 Results

Our results show a clear pattern: the baselines often produce realistic images, but reliability drops when the product must be used correctly. Overall quantitative scores are provided in Table 2.

Table 2: Average scores on a 0–10 rubric across the three evaluated baselines.

Model	Identity	Prompt	Realism	Affordance	Acceptable Rate
FireRed-Image-Edit	7.2	7.5	8.1	5.9	43%
LongCat-Image-Edit	6.5	6.2	8.5	5.6	37%
Nano Banana Pro	7.8	8.2	8.6	6.8	57%

Nano Banana Pro was strongest overall. FireRed was the stronger open-source baseline for identity, while LongCat often looked natural but missed product semantics more frequently.

4.2 Category-Level Observations

Different product categories presented unique challenges for the models. The specific acceptable rates and primary failure modes per category are detailed in Table 3.

Table 3: Category-level observations outlining acceptable rates and dominant editing issues.

Category	Acceptable Rate	Dominant Issue
Mobile/tablet	67% (over 6 tasks)	Screen/text drift
Appliance/UPS	60% (over 5 tasks)	Button/control alignment
Laptop accessory	40% (over 5 tasks)	Small text and key layout
Beauty tool	38% (over 4 tasks)	Extra hand / wrong use
Small electronics	33% (over 6 tasks)	Affordance ambiguity
Tripod/selfie stick	25% (over 4 tasks)	Levitation and scale

5 Discussion and Error Analysis

The project's central intuition is product identity and object affordance are deeply coupled. A generated image may score high on visual realism while still being entirely unusable because the product changes slightly, the hand contacts the wrong part of the item, or the scale becomes physically impossible.

This highlights why generic image quality metrics alone are insufficient for e-commerce catalog enrichment. The evaluation needs product-aware and affordance-aware checks that reflect whether the image would actually be acceptable in a professional catalog workflow.

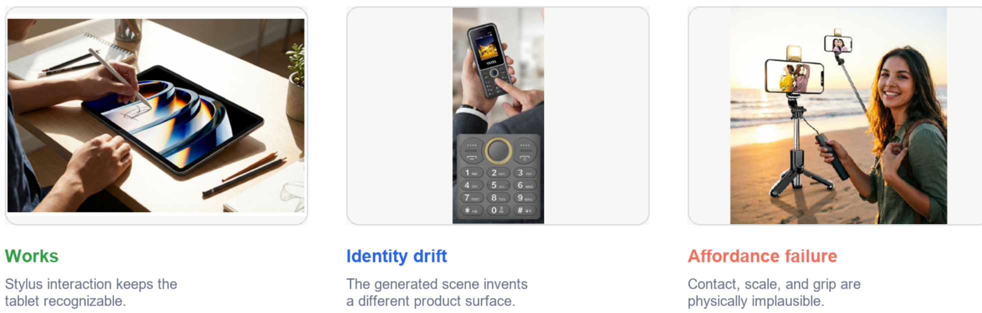

Figure 3: Qualitative examples highlighting the dichotomy between perceptual realism and affordance/identity failures.

6 Conclusion and Future Work

The project provides a coherent data formulation, baseline evaluation, results story, and next step method for multimodal guided image editing. The final takeaway is that an e-commerce editor should optimize for sellable correctness, not just visually pretty images. Future work to improve upon this includes:

- **Near-term improvement:** Acquire target lifestyle images from existing catalogs (e.g., Flipkart) to create stronger paired data.
- **Model improvement:** Add affordance conditioning, 3D pose awareness, or expert-routed loss functions.
- **Evaluation improvement:** Formalize *AffordScore* by comparing generated-image affordance maps against product-specific affordance priors.
- **Deployment improvement:** Build SOP-aligned checks for text/logo retention, color drift, hallucinated accessories, and interaction validity.

References

[1] Brooks, T., Holynski, A., and Efros, A. A. (2023) InstructPix2Pix. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).*

[2] Peebles, W., and Xie, S. (2023) Scalable Diffusion Transformers. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV).*

[3] Oquab, M., et al. (2023) DINOv2: Learning Robust Visual Features without Supervision. *arXiv preprint arXiv:2304.07193.*

[4] Rafailov, R., et al. (2023) Direct Preference Optimization. *arXiv preprint arXiv:2305.18290.*

[5] Ye, H., et al. (2023) IP-Adapter. *arXiv preprint arXiv:2308.06721.*

[6] Zhang, L., Rao, A., and Agrawala, M. (2023) ControlNet. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV).*